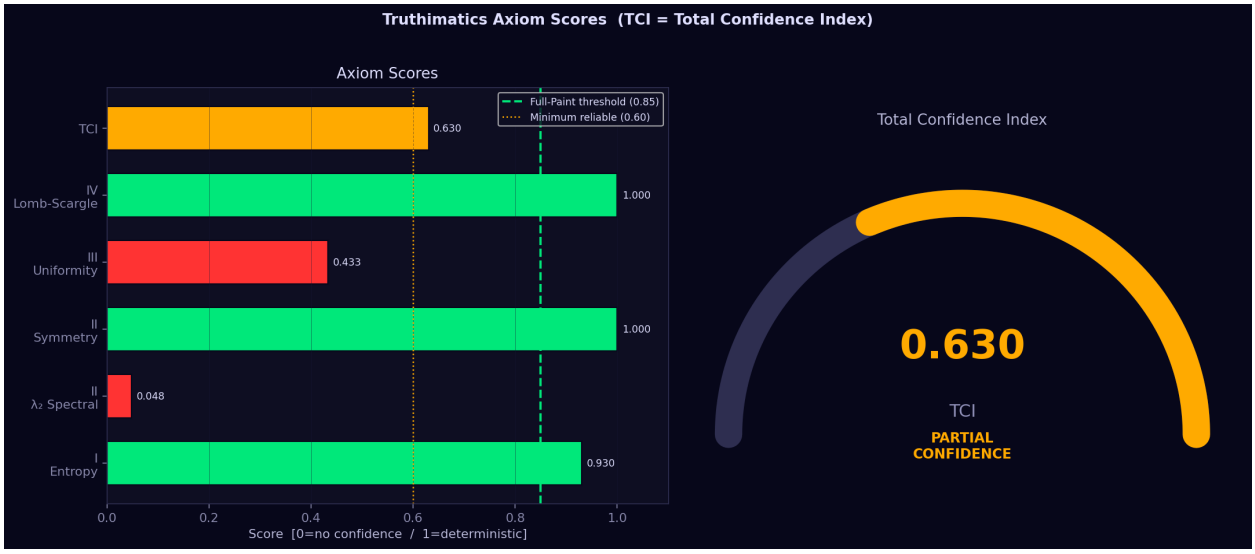
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_A6A016C549
Taxa	Therizinosaurus
Clade	Theropod
PBDB Records	2
Mass (regression)	4836.0 kg (Campione & Evans 2012)
Mass (volumetric)	2760.9 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	3.248 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6300 (partial)
Axiom I Entropy	0.9299
Axiom II λ■	0.1938 (symmetry 100%)
Axiom III Uniformity	0.4331
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	60.0% Measured / 40.0% Inferred
Elements	25
CoG (m)	[0.52, 0.936, 0.0]
Generated	2026-04-30 23:54 UTC

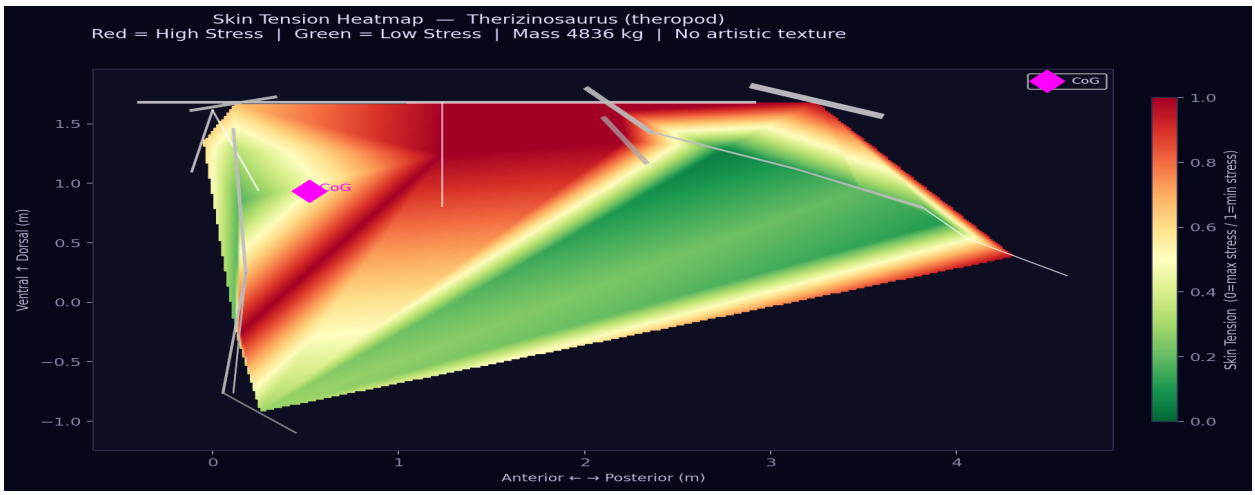
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



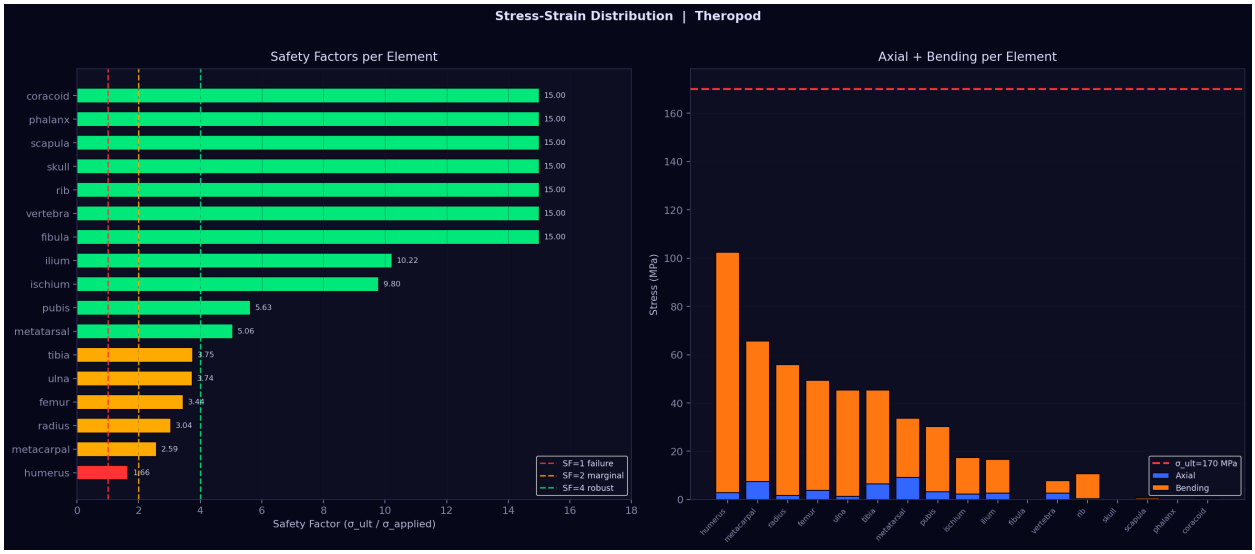
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

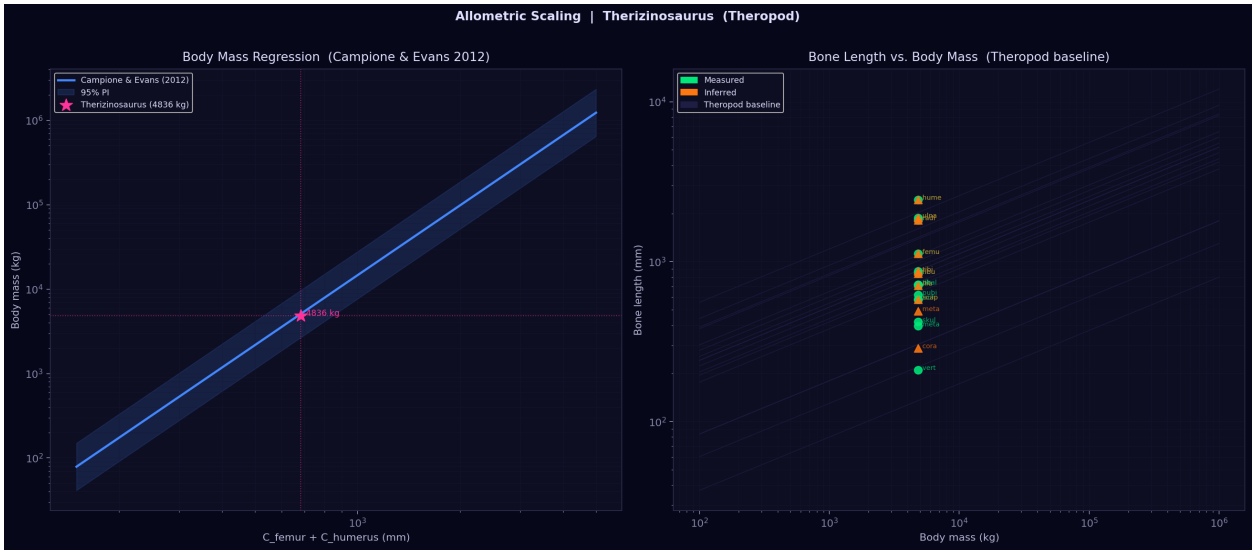


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.176	0.311	0.487	15.000	OK
femur	3.790	45.582	49.372	3.443	WARNING
fibula	0.000	0.000	0.000	15.000	OK
humerus	2.799	99.631	102.430	1.660	CRITICAL
ilium	2.704	13.927	16.631	10.222	OK
ischium	2.297	15.057	17.353	9.797	OK
metacarpal	7.357	58.381	65.738	2.586	WARNING
metatarsal	9.249	24.373	33.622	5.056	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	3.239	26.952	30.190	5.631	OK
radius	1.599	54.264	55.863	3.043	WARNING
rib	0.342	10.430	10.772	15.000	OK
scapula	0.164	0.567	0.730	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	6.562	38.717	45.278	3.755	WARNING
ulna	1.363	44.087	45.450	3.740	WARNING
vertebra	2.704	5.149	7.853	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	2.481 m	Femur + tibia + metatarsal
Estimated Speed	17.1 km/h (4.76 m/s)	Gait: Trot | Fr=0.932
Stride Length	5.572 m	Alexander (1976) / Thulborn (1990)
Bite Force	732.0 N (74.6 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	420.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.3506	Circumference / length (dimensionless)
Limb Strength Index	3.4e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	9.46 m	Allometric from femur length
Est. Neck Length	1.32 m	Allometric from femur length
Est. Tail Length	3.59 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	287.3	217.3	5435.49	6785.1	Constraint	0.40	15.00
femur	1120.0	125.0	7009.68	8750.2	Measured	0.88	3.44
femur_right	1120.0	125.0	7009.68	8750.2	Symmetry	0.84	3.44
fibula	840.0	38.0	342.96	359.7	Measured	0.88	15.00
fibula_right	840.0	38.0	342.96	359.7	Symmetry	0.84	15.00
humerus	2440.0	92.0	8272.29	10326.3	Measured	0.88	1.66
humerus_right	2440.0	92.0	8272.29	10326.3	Symmetry	0.84	1.66
ilium	710.0	148.0	6229.34	7776.1	Measured	0.88	10.22
ilium_right	710.0	148.0	6229.34	7776.1	Symmetry	0.84	10.22
ischium	580.0	95.0	2096.70	2617.3	Measured	0.88	9.80
metacarpal	395.0	55.0	178.31	146.7	Measured	0.88	2.59
metatarsal	490.9	80.0	1258.84	1571.4	Constraint	0.65	5.06
phalanx	720.0	42.0	153.22	118.6	Measured	0.88	15.00
pubis	620.0	80.0	1589.39	1984.0	Measured	0.88	5.63
radius	1820.0	72.0	3779.17	4717.5	Measured	0.88	3.04
radius_right	1820.0	72.0	3779.17	4717.5	Symmetry	0.84	3.04
rib	720.0	38.0	416.45	519.9	Measured	0.88	15.00
scapula	580.0	225.0	11761.24	14681.6	Measured	0.88	15.00
scapula_right	580.0	225.0	11761.24	14681.6	Symmetry	0.84	15.00
skull	420.0	320.0	17226.99	21504.4	Measured	0.88	15.00
tibia	870.0	95.0	3145.04	3926.0	Measured	0.88	3.75
tibia_right	870.0	95.0	3145.04	3926.0	Symmetry	0.84	3.75
ulna	1880.0	78.0	4581.49	5719.1	Measured	0.88	3.74
ulna_right	1880.0	78.0	4581.49	5719.1	Symmetry	0.84	3.74
vertebra	210.0	148.0	1842.48	2300.0	Measured	0.88	15.00